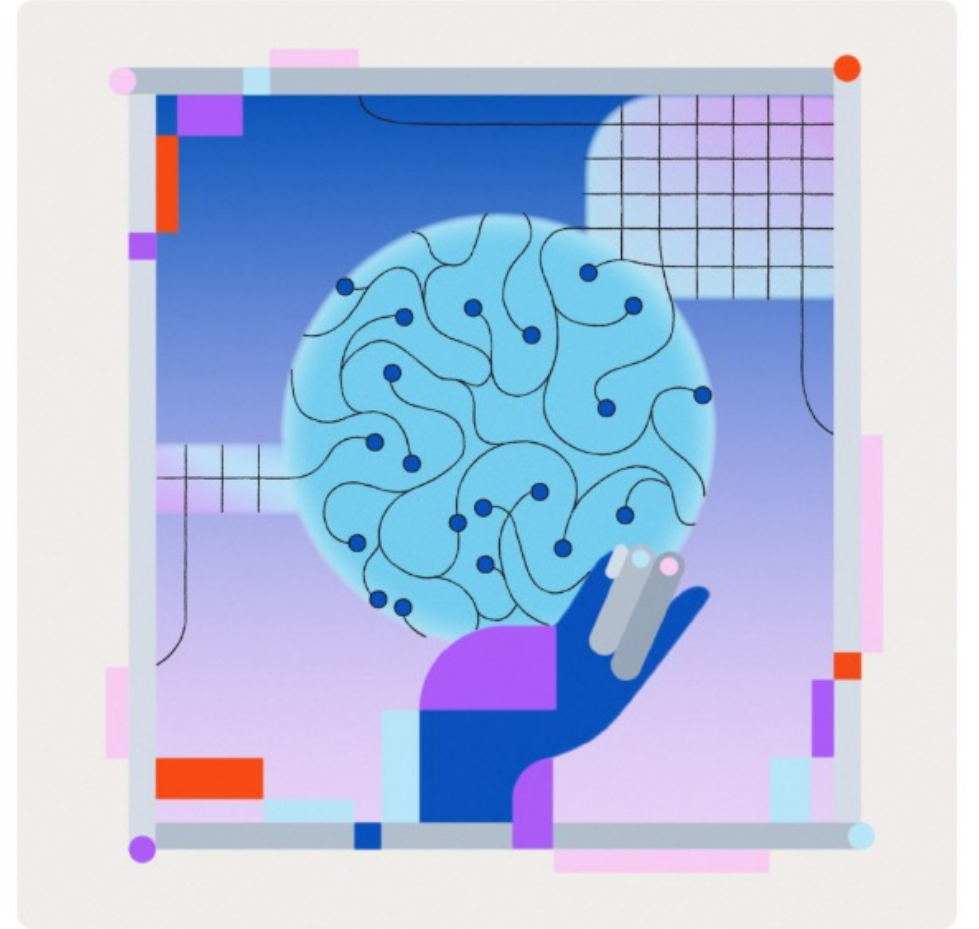


ML, AI and Gen AI



AI in 2026

- Democratized Gen AI
- LLMs everywhere
- Agents

AI

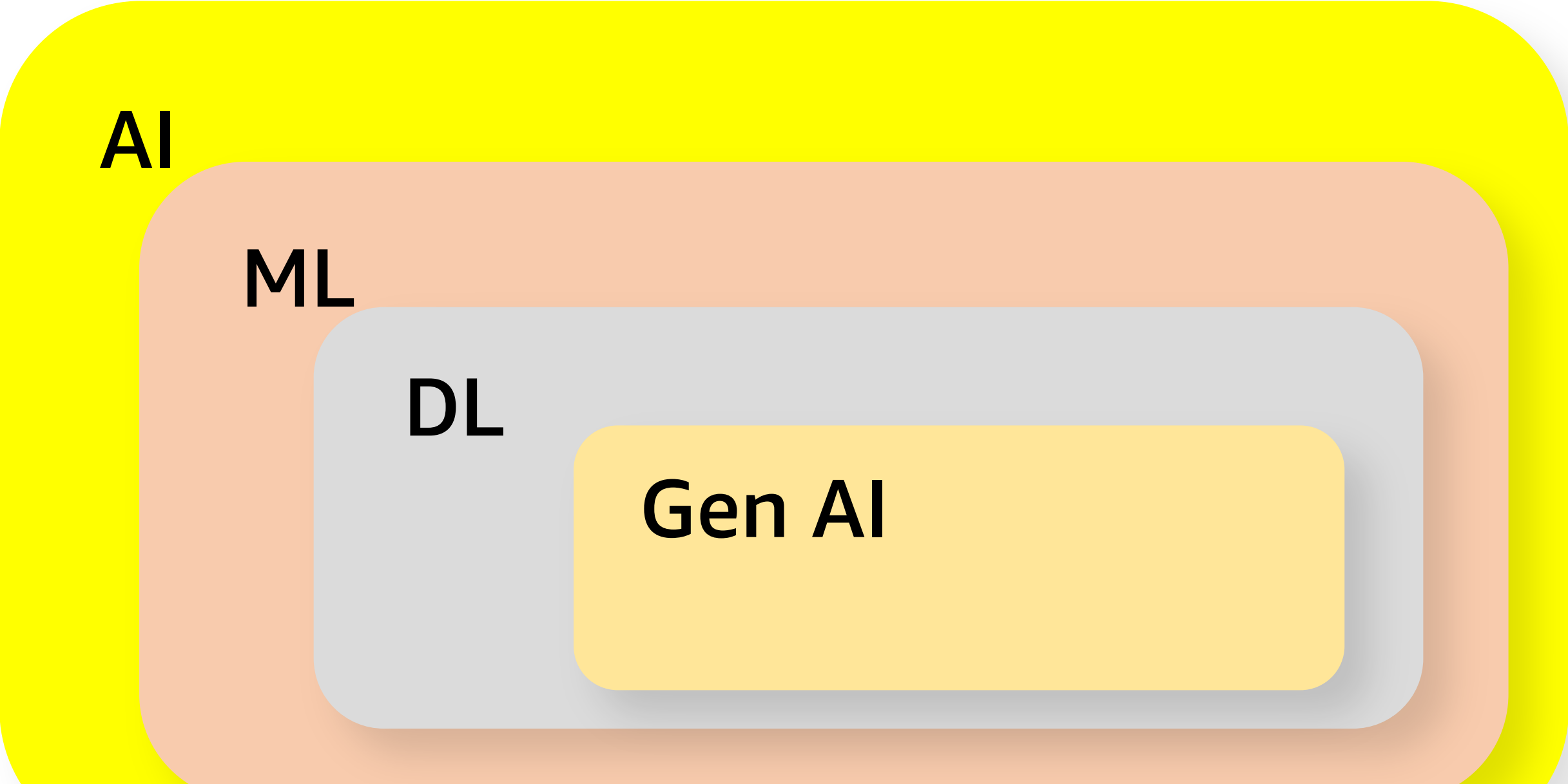
AI is the capability of **computational systems** to perform tasks typically associated with **human intelligence**:

Learning

Reasoning

Problem-solving

Decision-making



The diagram consists of four nested rounded rectangles. The outermost rectangle is yellow and contains the text 'AI'. Inside it is an orange rectangle containing 'ML'. Inside the orange rectangle is a light blue rectangle containing 'DL'. Finally, inside the light blue rectangle is a yellow rectangle containing 'Gen AI'. This visualizes that Gen AI is a subset of DL, which is a subset of ML, which is a subset of AI.

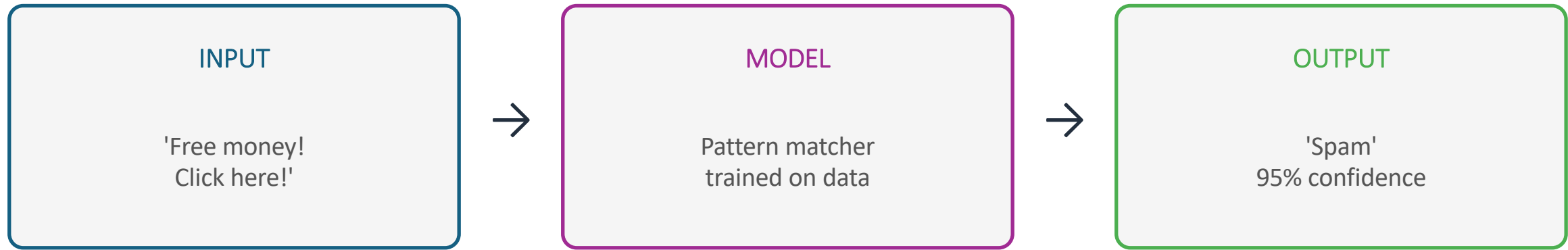
AI

ML

DL

Gen AI

How Machines Learn



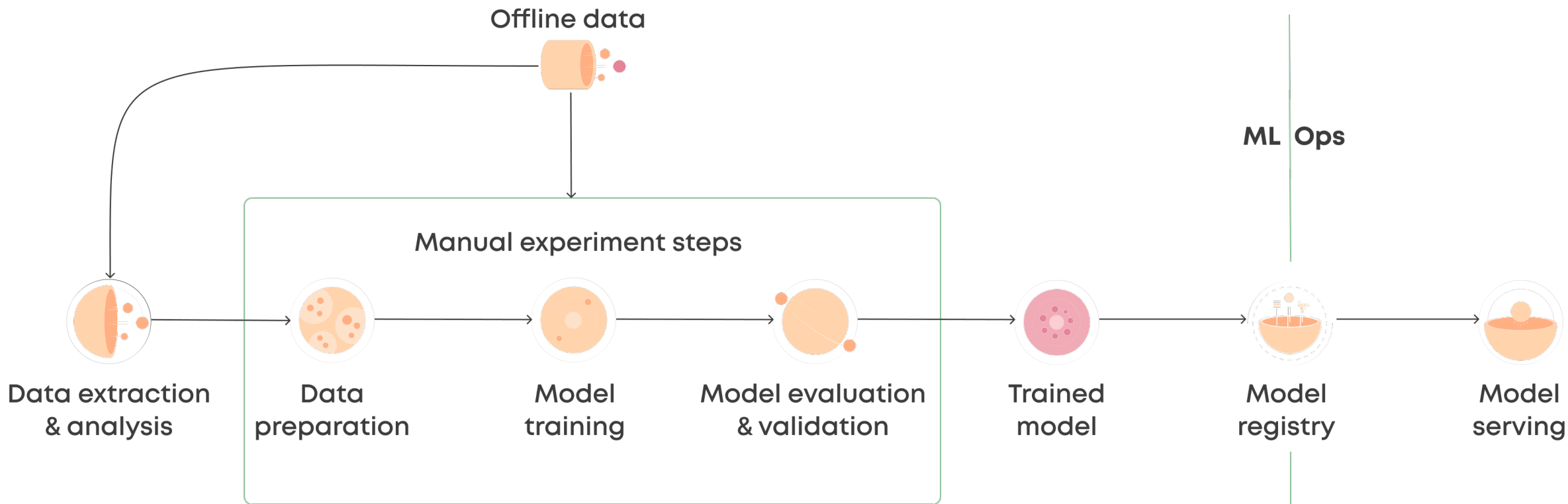
Training

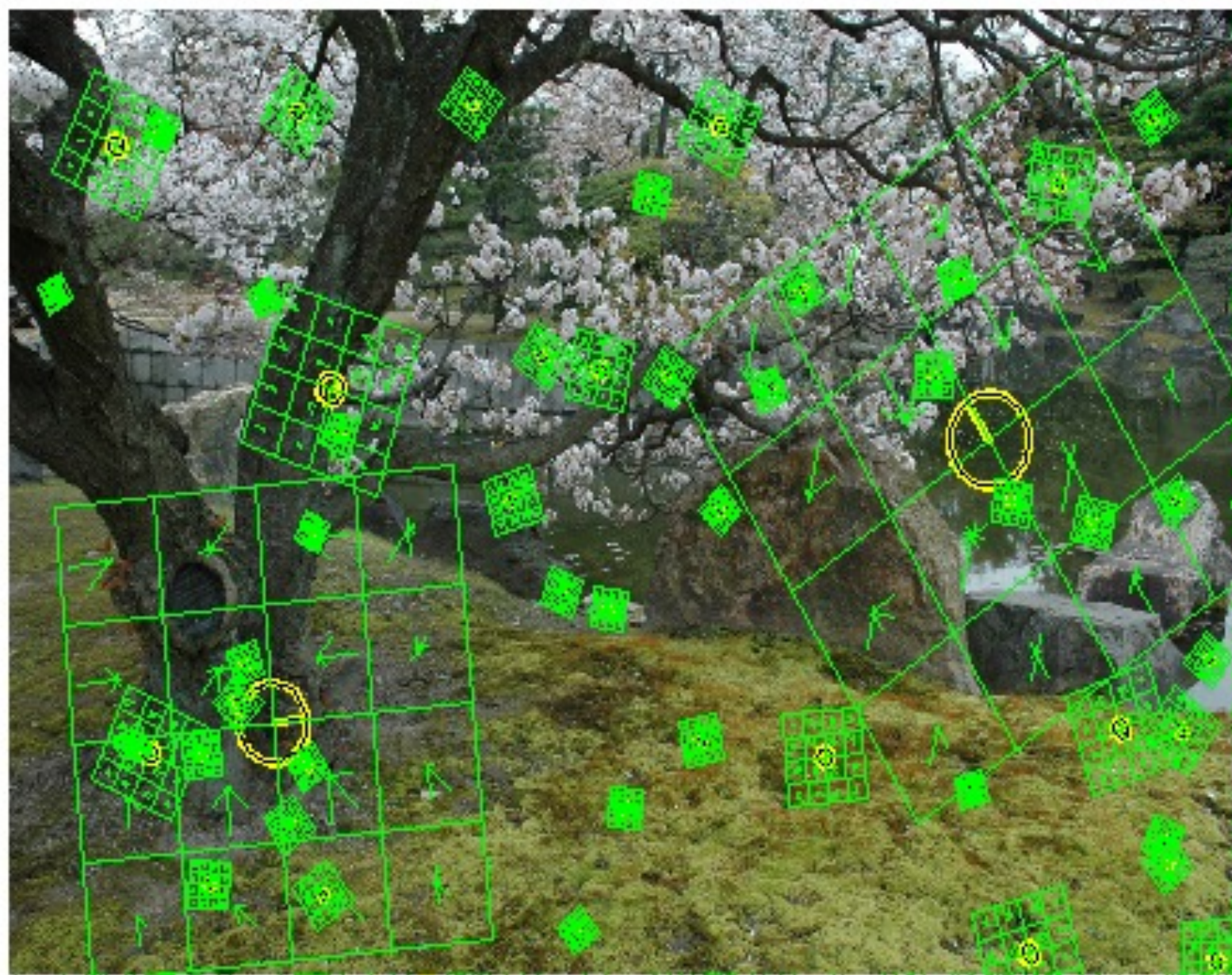
Show the model thousands of labeled examples so it learns the patterns

Inference

Model makes predictions on new, unseen data in real time

ML Loop





ML Paradigms



Supervised

Labeled data, learn input→output



Unsupervised

Find hidden structure, no labels



Semi-supervised

Small labeled + large unlabeled



Self-supervised

Model creates own labels



Reinforcement

Learn through rewards & penalties

Supervised Learning

Uses labeled datasets — each example has a known correct answer.

The model learns a mapping: inputs $\rightarrow$ outputs.

Classification

Discrete categories: spam detection, image recognition

Regression

Continuous values: price prediction, temperature forecasting

Unsupervised Learning

No labels — discovers hidden patterns and structures on its own.

Clustering

k-means, DBSCAN, hierarchical

Dimensionality Reduction

PCA, t-SNE, UMAP

Anomaly Detection

Outliers & unusual patterns

Semi & Self-supervised

Semi-supervised

Small labeled + large unlabeled dataset

Leverages structure in unlabeled data to improve predictions

Example: Label 100 images, use 10,000 unlabeled to boost accuracy

Self-supervised

Model creates its own labels from data structure

Masked language modeling, next-token prediction, contrastive learning

Foundation for BERT, GPT pre-training

Reinforcement Learning

Agent interacts with environment, maximizing cumulative reward through trial and error.

States & Actions

Rewards & Penalties

Policy & Value Fn

Explore vs Exploit

Applications:

AlphaGo · Robotics · Recommendations · RLHF

Classical Algorithms



Linear / Logistic Regression

Interpretable baselines



Support Vector Machines

High-dimensional spaces



Naive Bayes

Fast, great for text



Decision Trees & Random Forests

Best for tabular data



k-Nearest Neighbors

Instance-based learning



Gradient Boosting (XGBoost)

Competition winners

Evaluation Metrics

Classification

Accuracy

Precision & Recall

F1 Score

AUC-ROC

Confusion Matrix

Regression

MSE / RMSE

MAE

R^2 Score

Cross-validation

Adjusted R^2

Feature Engineering

Manual (Classical ML)

Domain expertise required
Feature importance analysis

Selection methods:

- Filter (correlation)
- Wrapper (RFE)
- Embedded (Lasso)

Automatic (Deep Learning)

Learned representations
Hierarchical extraction

Tradeoffs:

- Requires more data
- Less interpretable
- More compute needed

Deep Learning

Multi-layered neural networks — automatic hierarchical representations.



CNNs

Computer vision, image recognition



RNNs / LSTMs

Sequential data, time series



Autoencoders

Compression, anomaly detection

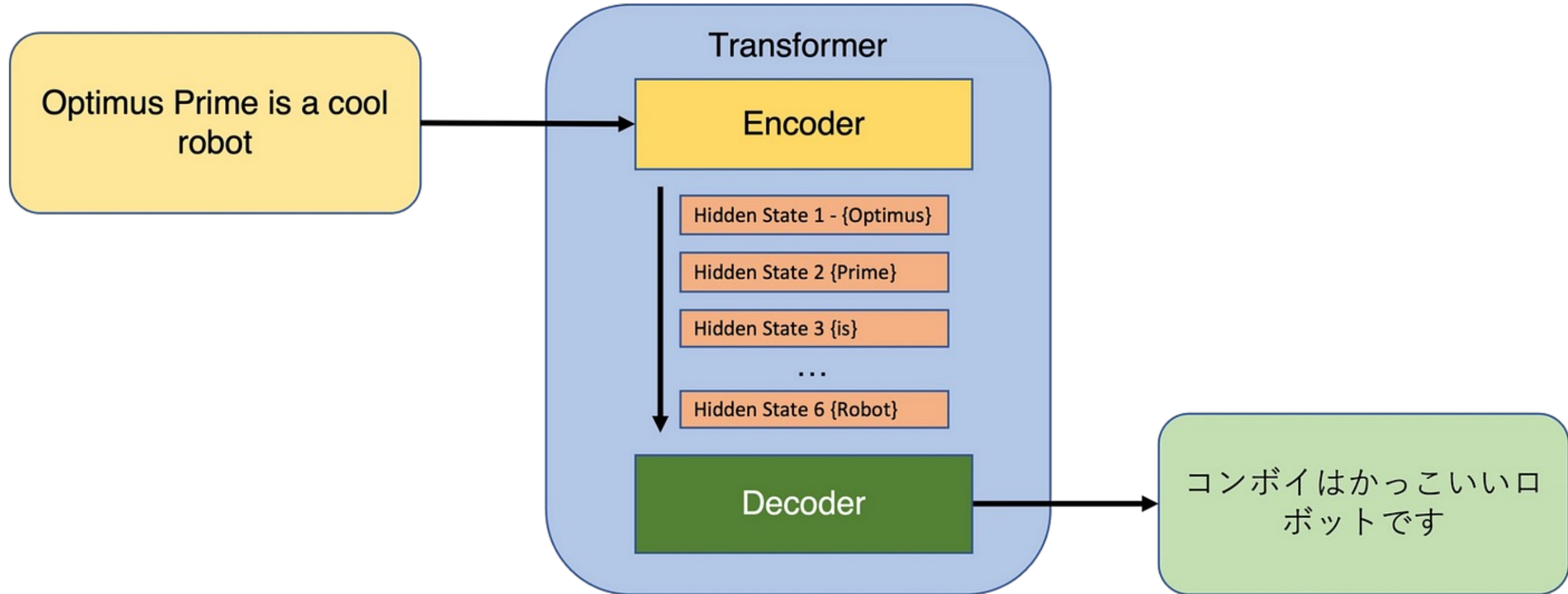


GANs

Generative adversarial networks



Transformer



Transformer

Eliminated sequential processing through self-attention.
Each token attends to all others simultaneously.

Self-Attention: relate all parts in parallel

Parallelizable: no sequential dependency

Encoder-Decoder: understand + generate

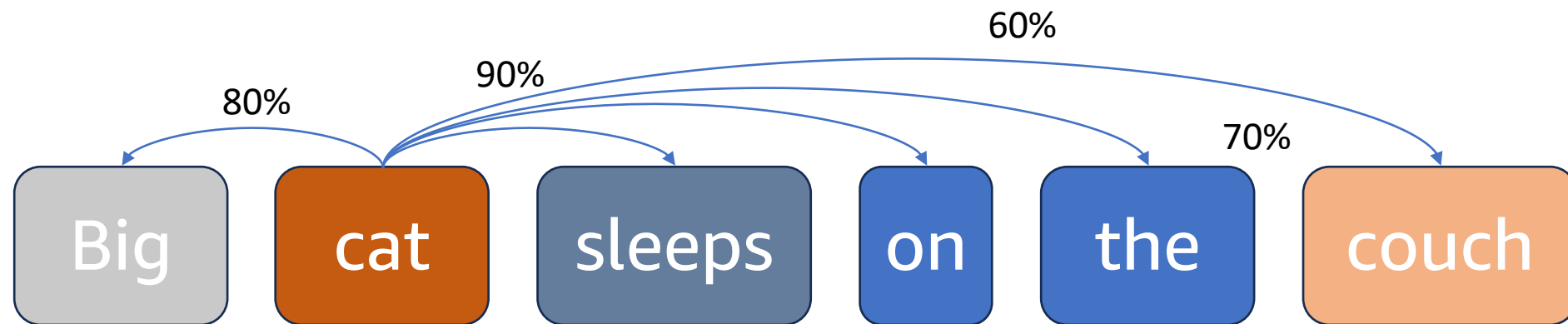
Scalable: foundation for all modern LLMs

Attention

$$Q \times K^T / \sqrt{d}$$

softmax $\rightarrow$ V

Attention



Completion

Big

cat

sleeps

on

the

couch

Completion

Big cat sleeps on the sofa 95%

Completion

83%

Big

cat

sleeps

on

the

tree

Completion

Big cat sleeps on the cloud

11%

Completion

Big

cat

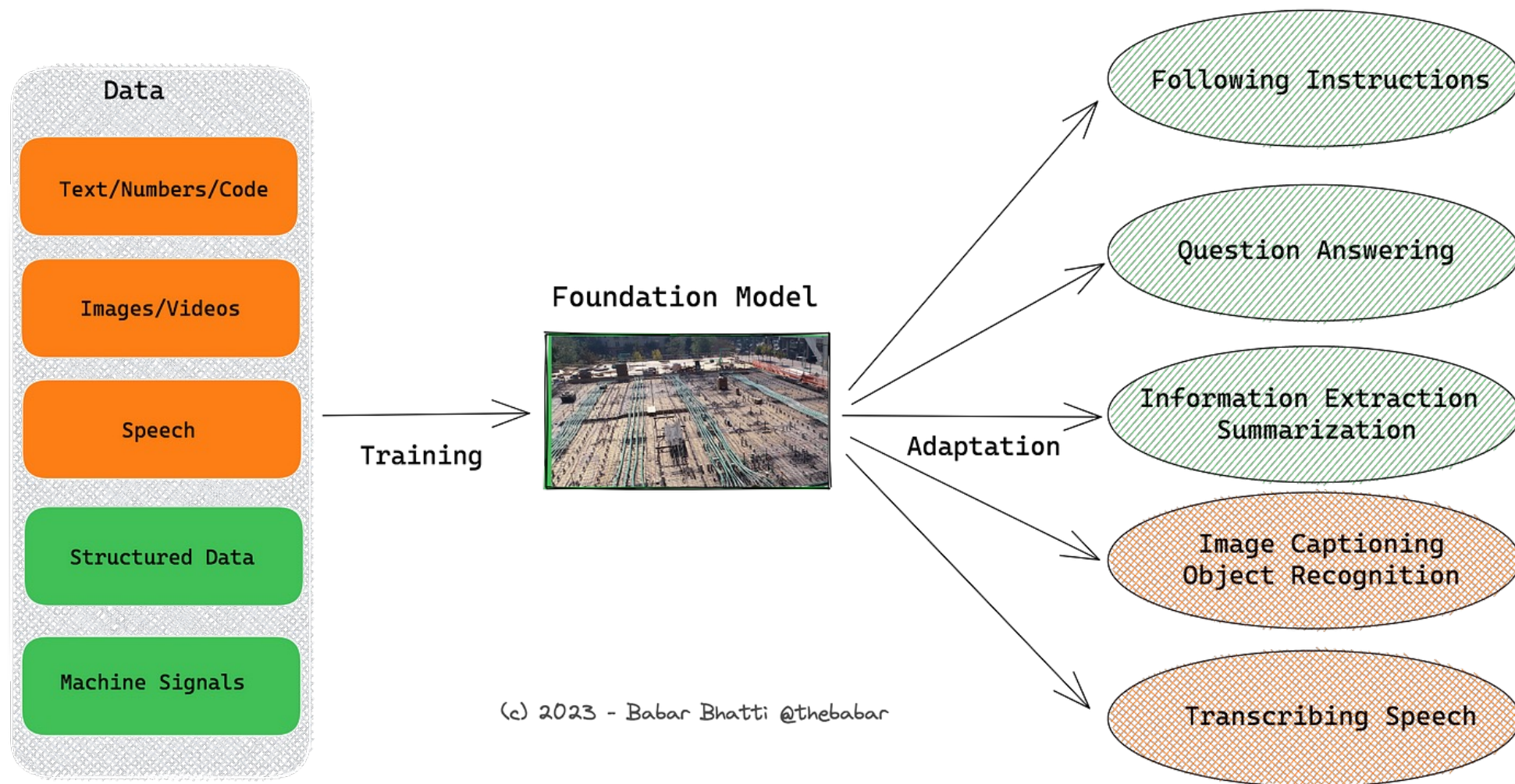
sleeps

on

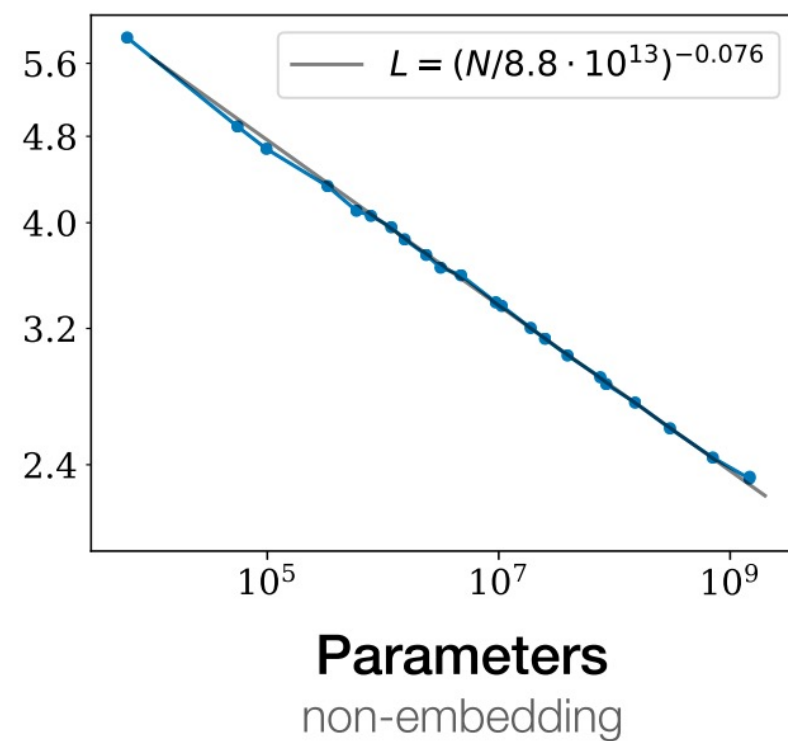
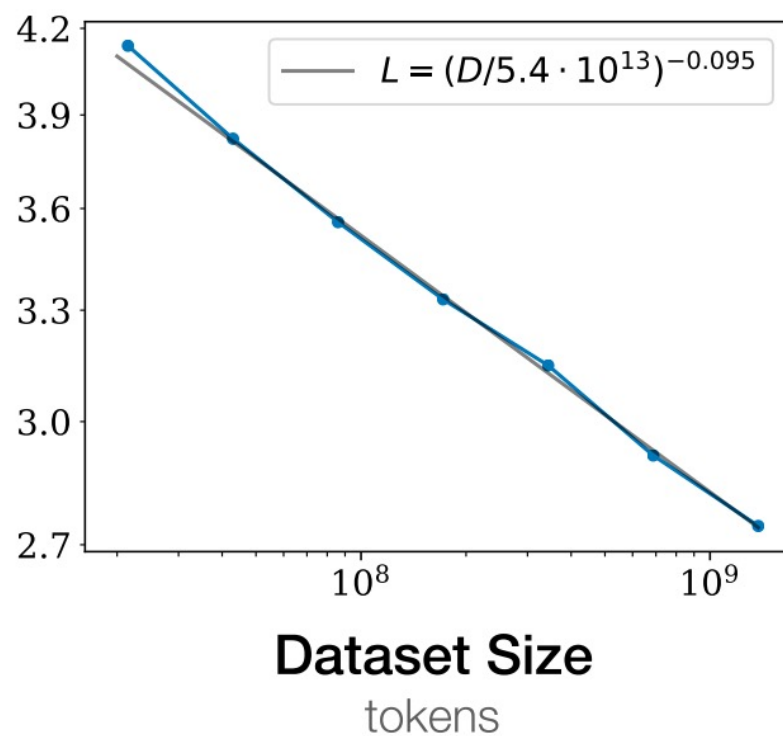
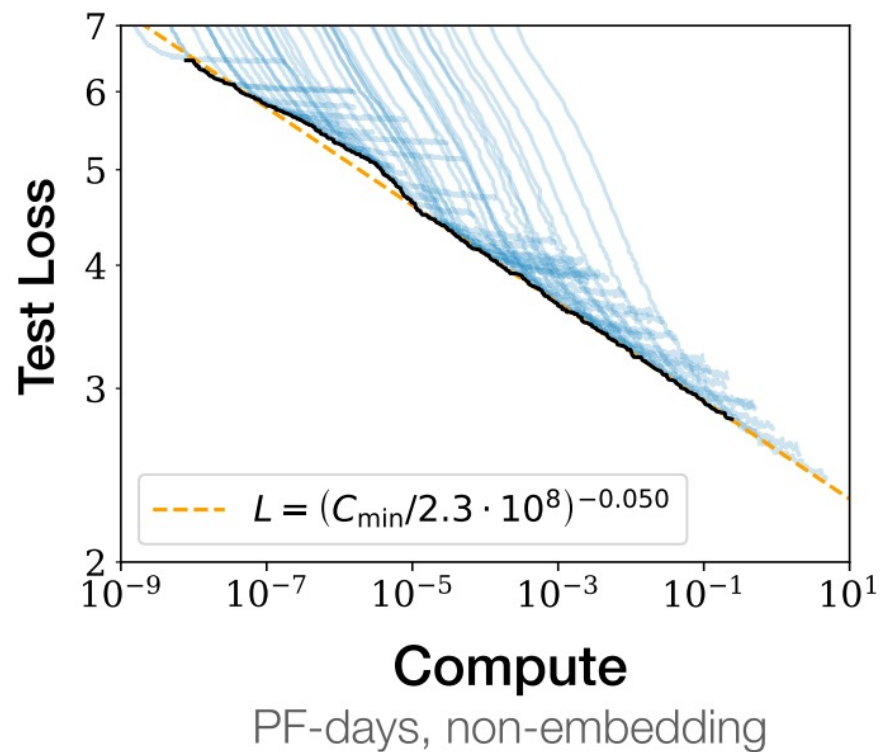
the

couch

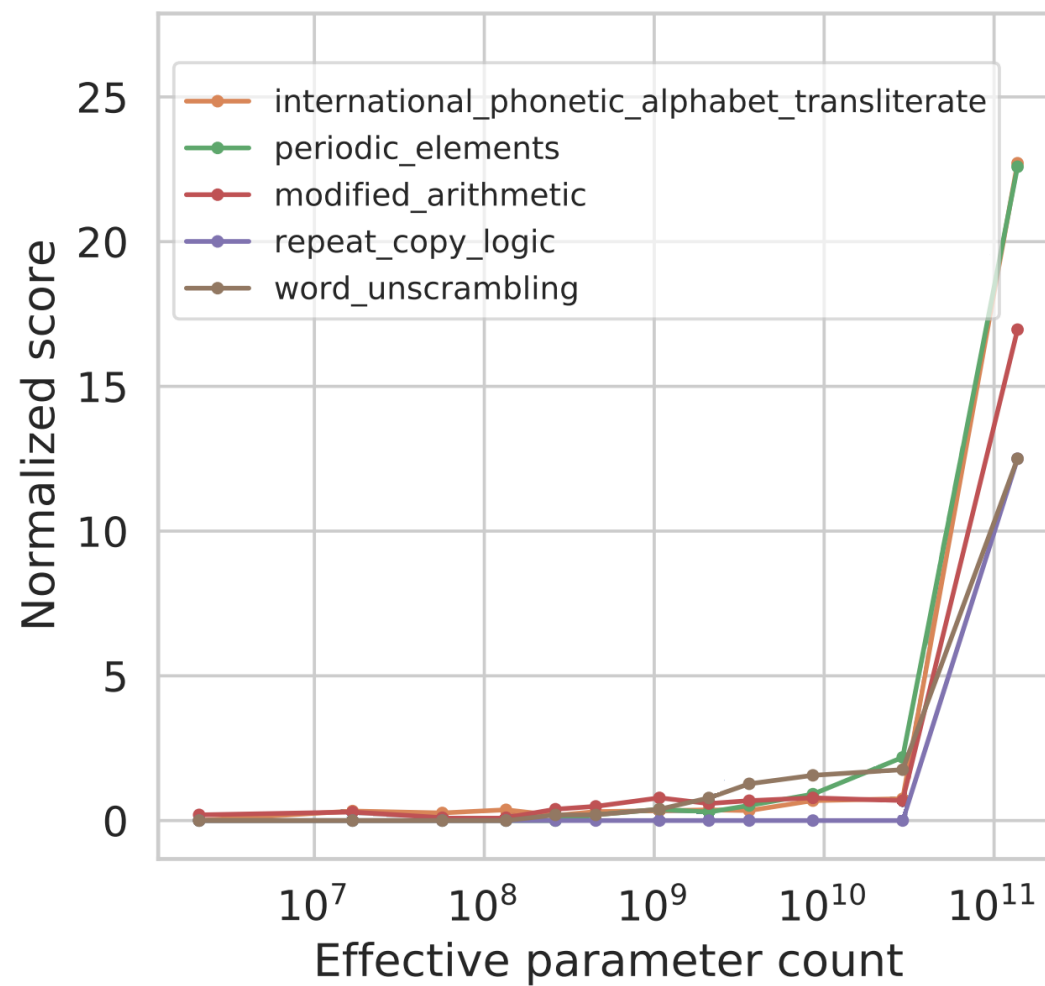
FMs



Scaling?



Abilities



Summary

Classical ML → Deep Learning → GenAI: evolution, not replacement

Choose the right paradigm for the problem

Supervised learning still powers most production systems

GenAI is the frontier, but fundamentals matter

Questions?